

Algebra

Topic 5: Substitution

QUESTION 1 If $a = 3$, find the value of the following expressions.

- a $5a =$ _____ b $8a + 1 =$ _____ c $4a - 1 =$ _____
d $a^2 =$ _____ e $3a^2 =$ _____ f $(a + 1)(a - 1) =$ _____
g $8a - 10 =$ _____ h $a^2 - a =$ _____ i $a \div 6 =$ _____
j $9a \div 3 =$ _____ k $27 \div a =$ _____ l $(a - 2) \div 9 =$ _____

QUESTION 2 If $a = 2$, $b = 5$, $c = 7$ and $d = 9$, find the value of

- a $a + b =$ _____ b $c + d =$ _____ c $ab + cd =$ _____
d $abc =$ _____ e $2a + 3b =$ _____ f $2b + 5c =$ _____
g $9ab =$ _____ h $7b - c =$ _____ i $8b \div a =$ _____
j $3a - a^2 =$ _____ k $a + b + c =$ _____ l $8c - a =$ _____

QUESTION 3 Complete the following.

- a $b = a + 4$

a	1	2	3	4	5
b					

b $y = 2 + 5x$

x	0	1	5	7
y				

c $m = 12 - l$

l	0	1	5	7	9
m					

d $y = 3x - 1$

x	1	2	3	4	5	6
y						

e $q = 4p + 3$

p	0	1	2	3	4
q					

f $v = 20 - 3u$

u	1	2	3	4	5	6
v						

QUESTION 4 Complete the following table.

	m	n	$m + n$	$m - n$	mn	m^2	n^2	$3m + 2n$
a	4	2						
b	3	1						
c	7	2						
d	6	1						
e	9	3						
f	8	7						

Algebra

Topic 10: One-step equations — addition and subtraction

QUESTION 1 Solve the following one-step equations.

a $x + 9 = 14$

b $a + 6 = 35$

c $m - 6 = 5$

d $m + 8 = 34$

e $x + 4 = 14$

f $n - 7 = 12$

g $p - 3 = 16$

h $m + 5 = 19$

i $y + 8 = 22$

j $y - 5 = 12$

k $t - 6 = 2$

l $a - 7 = 16$

m $x - 7 = 28$

n $y - 7 = 4$

o $8 + x = 24$

QUESTION 2 Solve the following equations.

a $a + 16 = 34$

b $y + 7 = 13$

c $6 + y = 18$

d $15 + n = 16$

e $x - 20 = 32$

f $x - 2 = 14$

g $x + 12 = 38$

h $y - 5 = 29$

i $a + 7 = 15$

j $m - 6 = 15$

k $m - 10 = 39$

l $x - 71 = 89$

m $y - 6 = 9$

n $t - 7 = 16$

o $12 + a = 16$
